

REVIEW - JANUARY 2025

1. Let $f(x) = \begin{cases} e^{1/(x+2)} & \text{if } x < -2 \\ c(1+x)^{2/3} + \cosh(x+2) & \text{if } x \geq -2. \end{cases}$

- a) Decide if there exists c such that f is differentiable at $x = -2$.
- b) Determine the domain of f' .

2. Define $\int_0^{(x-2)^2} e^t(t-4)dt$.

Find the critical points of F and decide if they are local maximum or minimum.

3. Find the area bounded by the curves:

a) $y = \tan^3(x) \sec^5(x)$ and the x -axis on $[0, \frac{\pi}{6}]$.

b) $y = \ln(x) (2x^2 + 4\sqrt{x})$ and the x -axis on $[1, 4]$.

4. Let $g(x) = (1 + \frac{\pi}{4})x - \frac{1}{2}x^2$ and $h(x) = x \arctan(x) - \frac{1}{2} \ln(x^2 + 1)$. Find the values of $x \in (0, +\infty)$ for which f reaches a local or absolute extreme value if

$$f(x) = \int_0^{g(x)-h(x)} |\tanh t| dt$$

5. Find the volume of the solid obtained by the rotation about the x -axis of the area enclosed by the curve $y = \sqrt{x} \tan(x)$ and the x -axis on $[0, \frac{\pi}{3}]$.

6. Find the length of the parabola $y = x^2$ between the points $(0, 0)$ and $(2, 4)$.

7. Find the area of the surface generated by the rotation of the arc of the curve $y = e^x$ from $x = 0$ to $x = 1$, around the x -axis.

8. Find the area of the surface generated by the rotation of the arc of the parabola $y = 3x^2 + 1$ for $0 \leq x \leq 1$, around the y -axis.

9. Compute the limit

$$\lim_{x \rightarrow \pi} \frac{8 \cos(x) + 8 - 4(x - \pi)^2 + (x - \pi)^4/3}{(x - \pi)^6}$$

10. Find all the values of x such that the series is convergent

$$a) f(x) = \sum_{n=1}^{\infty} \frac{(-2)^n (2x+5)^n}{\sqrt{3n-1}}$$

$$b) g(x) = \sum_{n=3}^{\infty} \left[\frac{(2x+8)^n}{(n-2)} + \frac{(x+4)^n}{n^2} \right]$$

$$c) h(x) = \sum_{n=0}^{\infty} \left[\frac{2x+3}{2x-1} \right]^n$$

What is the value of $h(x)$ for each x such that the series is convergent?

11. Use any test to determine whether the series is absolutely convergent, conditionally convergent, or divergent:

a) $\sum_{n=3}^{\infty} \left(\frac{\ln n}{n} \right)^2$

b) $\sum_{n=3}^{\infty} \frac{\pi^n}{2^{2n-1}}$

c) $\sum_{n=1}^{\infty} \ln \left(\frac{n}{n+1} \right)$

12. Find a rational number whose distance from $\frac{1}{\sqrt{4.4}}$ is at most $\frac{1}{2000} = \frac{5}{10^4}$.

13. Determine if the following series is convergent or divergent $\sum_{n=1}^{\infty} a_n$, where $a_1 = 1$ and for all $n \in \mathbb{N}$, $a_{n+1} = \left(\frac{n-1}{n+2}\right)^{\sin(n)/4} a_n$.

14. Use geometric series to find the Taylor series of the given functions centered at a . Find the interval of convergence in each case:

a) $f(x) = \frac{x^2 + 4x + 4}{x}$ at $a = 2$, b) $g(x) = -\frac{1}{x^2}$ at $a = 2$, b) $h(x) = \cosh(x)$ at $a = \ln(2)$.

15. Find all the $x \in \mathbb{R}$ such that the series is convergent and the result of the infinite sum, when possible:

a) $\sum_{n=1}^{\infty} \left(\frac{1}{8-x^2} \right)^n$

b) $\sum_{n=4}^{\infty} e^{-2nx}$

16. Find an approximation of $\int_0^1 e^{-x^2} dx$ with an accuracy of 10^{-5} .